

Hazard Assessment Form

Version Control

Version	Date	Author	Notes	Approved By:	Date:
1	N/A	Wendy	New Form Created		
2	16-Dec-2025	Megha	Added extra columns Generic Information Section Revised Formatting and fillable PDF conversion		
3	30-Jan-2026	Sandra	First Version of Hazards for Maxim Mechanical Group Inc.	Peter Godler	Feb 4, 2026

Hazard Assessment Form

Date of Assessment:	January 30, 2026	Job/position:	Welder	Date Reviewed:	Feb 11, 2026
Assessment performed by:	Sandra Tutka	Supervisor:	BJ Wilson	Date Implemented:	Feb 25, 2026

		Inherent Risk Before Controls			Hierarchy of Control in Place	Residual Risk After Controls		
Task	Hazards Present	Likelihood	Severity	Risk Level		Likelihood	Severity	Risk Level
Arc welding (SMAW, MIG, TIG, FCAW)	UV/IR radiation, molten metal, burns, arc eye, skin damage	Expected (4)	Catastrophic (4)	16	Engineering: Welding screens/curtains. Administrative: Restricted welding areas, training. PPE: Welding helmet (proper shade), FR clothing, gloves.	Possible (2)	Moderate (2)	4
Welding fumes exposure	Metal fumes, Respiratory irritation, long-term lung disease	Probable (3)	Catastrophic (4)	12	Substitution: Use lower-fume materials. Engineering: Local exhaust ventilation, fume extractors. Administrative: Exposure monitoring, training. PPE: Respirator	Possible (2)	Minimal (1)	2
Hot work in active industrial areas	Fire, explosion from flammables, burns, fire	Probable (3)	Catastrophic (4)	12	Engineering: Fire blankets, spark containment. Administrative: Hot work permit, fire watch, area inspection. PPE: FR clothing, gloves, face shield.	Possible (2)	Minimal (1)	2
Grinding welds / prep work	Flying particles, eye injuries, cuts, burns, hearing loss	Probable (3)	Moderate (2)	6	Engineering: Machine guards, spark containment. PPE: Safety glasses + face shield, cut-resistant gloves, hearing protection.	Possible (2)	Minimal (1)	2
Confined space work (vaults, boiler rooms, tanks)	Low oxygen, toxic gases, Asphyxiation, poisoning, fatality	Probable (3)	Catastrophic (4)	12	Elimination: Use external access methods if possible. Engineering: Ventilation systems, gas monitors. Administrative: Confined space program, entry permits, rescue plan.	Possible (2)	Moderate (2)	4
Working at heights (Lifts)	Falls from height, falling tools/materials	Probable (3)	Catastrophic (4)	12	Engineering: Guardrails, lift platforms. Administrative: Fall protection plan, ladder safety training. PPE: Fall arrest harness, hard hat.	Possible (2)	Minimal (1)	2

Hazard Assessment Form

Handling compressed gas cylinders	Cylinder tipping, valve damage, leaks, explosion	Probable (3)	Serious (3)	9	Engineering: Cylinder carts, caps, secure storage racks. Administrative: training, segregation of gases. PPE: Gloves, safety boots.	Remote (1)	Minimal (1)	1
Electrical shock from welding equipment	Faulty leads, wet conditions, Electric shock, burns	Probable (3)	Serious (3)	9	Engineering: Proper grounding. Administrative: Equipment inspections, lockout/tagout. PPE: Dry gloves, dielectric footwear	Possible (2)	Minimal (1)	2
Heat stress (especially in PPE or enclosed areas)	Heat exhaustion, dehydration	Possible (2)	Serious (3)	6	Engineering: Ventilation, cooled rest areas. Administrative: Work-rest cycles, hydration program, cooling stations. PPE: Cooling PPE	Possible (2)	Minimal (1)	2
Exposure to industrial noise	Noise-induced hearing loss	Possible (2)	Serious (3)	6	Engineering: Noise-dampening barriers. PPE: Hearing protection.	Possible (2)	Minimal (1)	2
Manual handling of steel components	Back strain, crushed fingers/toes	Possible (2)	Serious (3)	6	Engineering: Hoists, positioners, carts. Administrative: Team lifts, lift planning. PPE: Gloves, steel-toe boots.	Possible (2)	Minimal (1)	2
Slips, trips, and falls on job sites	Uneven ground, debris, wet surfaces	Expected (4)	Moderate (2)	8	Engineering: cable management. Administrative: Good housekeeping, site orientation. PPE: Slip-resistant, steel-toe boots.	Possible (2)	Minimal (1)	2
Exposure to coatings or contaminants (paint, galvanizing, oil residues)	Metal fume fever, respiratory illness	Possible (2)	Serious (3)	6	Elimination: Remove coatings before welding. Engineering: Local exhaust ventilation. Administrative: Hazard assessments, MSDS. PPE: Respirator as required.	Possible (2)	Moderate (2)	4
Interaction with other trades / mobile equipment	Struck-by moving equipment, dropped materials, serious injury, fatality	Probable (3)	Catastrophic (4)	12	Engineering: Barriers, defined walkways. Administrative: Site safety meetings, traffic control plans, hoisting plans, orientation. PPE: High-visibility clothing, hard hat, safety boots.	Probable (3)	Moderate (2)	6

Hazard Assessment Form

Exposure to opioids or other controlled substances	accidental contact, inhalation, or ingestion which may result in overdose, respiratory depression, or other serious health effects	Possible (2)	Catastrophic (4)	8	Elimination: Avoid entering high-risk environments. Engineering: Use barriers, proper ventilation. Administrative: Training on hazard recognition, safe handling procedures, incident response, naloxone training PPE: gloves, masks	Remote (1)	Moderate (2)	4
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Hazard Assessment Form

Level of Risk Chart

Likelihood: The likelihood parameter looks at the chance of a negative situation occurring.

Severity: Severity measures the degree of harm that exposure to the hazard could result in.

The level of risk is determined by the equation: **Likelihood x Severity = Risk** as per the table below.

Risk Chart		Likelihood of Occurrence			
		Expected (4) Very likely to happen	Probable (3) Could happen	Possible (2) A chance it could happen	Remote (1) Very unlikely to happen
Severity	Catastrophic (4) Death, critical injury, property damage over \$10K	16	12	8	4
	Serious (3) Medical Aid, lost time, property damage over \$5K	12	9	6	3
	Moderate (2) First Aid, Minor illness, Property damage over \$1K	8	6	4	2
	Minimal (1) No injury, minor damage	4	3	2	1

The risk score is then used to determine a risk category which will dictate actions required to manage the risk.

Risk Score Level	Risk Category	Action Required
16	Severe	Stop work immediately, implement controls to reduce the risk
9-12	High	Consider additional controls to reduce the risk. Create a Safe Work (Job) Procedure
4 -8	Medium	Consider additional controls, Create a Safe Work Practice
1 - 3	Low	Minimal risk, monitor the operation

Hierarchy of Controls

Most
effective



Least
effective

